

TrES-2b

R-0108

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_171005 · created 2026-07-16T03:50:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458031.67729 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.042
Transit depth (Rp/Rs)^2	2.18 +/- 1.24 [%]
Orbital Inclination (inc)	83.5 +/- 2.6
Airmass coefficient 1 (a1)	1.016 +/- 0.016
Airmass coefficient 2 (a2)	-0.011 +/- 0.048
Scatter in the residuals of the lightcurve fit is	1.56 %
Best Comparison Star	#3 - [312, 129]
Optimal Aperture	4.38
Optimal Annulus	5.32
Transit Duration (day)	0.055 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.042 vs 0.1254 (deviation 0.46 σ)
Duration fit vs published	0.055 d vs 0.0746 d (deviation 0.44 σ)
Mid-time O-C	0.5 min
Depth SNR	3.0
Residual scatter	1.56 %

Light curve

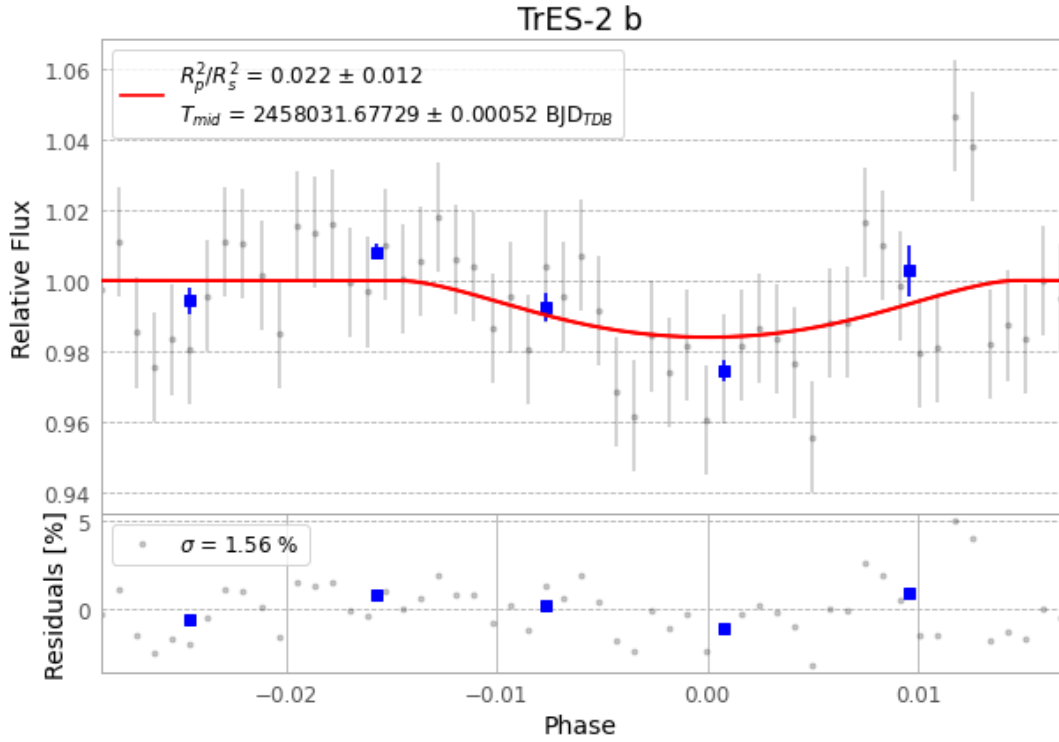


Figure 1 — FinalLightCurve_TrES-2 b_2017-10-04.png (SHA-256 908377a3e9745e7a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [216, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c690009b8125d1b4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

71 FITS frames (47 MB), TRES-2171005023012.FITS → TRES-2171005060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-171005.log (c691e4c28cf2...), FinalLightCurve_TrES-2 b_2017-10-04.png (908377a3e974...), FinalLightCurve_TrES-2 b_2017-10-04.csv (b2aba6390c90...)

Compute resources

Wall time 440.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:50Z.